

Міністерство освіти і науки України
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Проектування розподілених систем

Лабораторна робота 4
«Мікросервіси з використанням Messaging queue»

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Lab 4 - Мікросервиси з використанням Messaging queue

Завдання базується на функціоналі розробленому у попередньому завданні, і є його подальшим розвитком.

До **messages-service** необхідно додати:

- у якості каналу доставки повідомлень між **facade-service** та **messages-service** використати message queue
- можливість запускати одночасно декілька копій **messages-service**
- facade-service** випадковим чином обирає до якої копії **messages-service** звертатись для читання повідомлень

Пропонується використати у якості Messaging queue

- Kafka (10 балів) - <https://hub.docker.com/r/apache/kafka>

Функціональність системи

Клієнт взаємодіє з **facade-service** через HTTP POST та GET запити. У якості клієнта може бути *curl*, *Postman*, браузер у Dev mode та інше

- Необхідно розгорнути кластер **Messaging queue** - Hazelcast Distributed Queue (достаньо 2х серверів) чи Kafka (необхідно буде 3 сервери) та налаштувати реплікації черги (див. далі)
- При надходженні POST-запиту до **facade-service** він має додавати повідомлення з запиту до черги повідомлень
- Копії **messages-service** мають вичитувати повідомлення (схема - producer/consumer) та зберегти їх у пам'яті
- при GET запиті **facade-service**, з яким взаємодіє клієнт, випадковим чином обирає екземпляр **messages-service** та повертає по REST (HTTP) протоколу всі повідомлення, які зберігаються на даній копії

Завдання

- Запустити три екземпляри **logging-service** (локально їх можна запустити на різних портах), відповідно мають запуститись також три екземпляри Hazelcast
- Запустити два екземпляри **messages-service** (локально їх можна запустити на різних портах)

☐	Name	Container ID	Image	Port(s)	CPU (%)	Last started	Actions
☐	task4	-	-	-	27.92%	43 minutes ago	
☐	hazelcast3	94fb78964217	hazelcast/hazelcast:5.3	5703:5701	5.48%	46 minutes ago	
☐	hazelcast2	af80c9803d0d	hazelcast/hazelcast:5.3	5702:5701	5.83%	46 minutes ago	
☐	hazelcast1	0f95056a1c80	hazelcast/hazelcast:5.3	5701:5701	2.77%	46 minutes ago	
☐	zookeeper	15d788b4da05	confluentinc/cp-zookeeper	2181:2181	0.15%	46 minutes ago	
☐	kafka3	986de52c0b97	confluentinc/cp-kafka:7.3	9094:9094	2.89%	43 minutes ago	
☐	kafka2	b6907f4026ac	confluentinc/cp-kafka:7.3	9093:9093	2.27%	46 minutes ago	
☐	kafka1	0aac83842a80	confluentinc/cp-kafka:7.3	9092:9092	2.57%	46 minutes ago	

Showing 15 items

☐	☒	logging2	a84c0191cd9	task4-logging2	50052:50052	1.01%	43 minutes ago	☒	:	☒
☐	☒	logging3	947ad28fccb4	task4-logging3	50053:50053	0.99%	43 minutes ago	☒	:	☒
☐	☒	messages1	02c5f2fd9b59	task4-messages1	8882:8882	0.54%	43 minutes ago	☒	:	☒
☐	☒	logging1	4aab87cb2b42	task4-logging1	50051:50051	0.95%	43 minutes ago	☒	:	☒
☐	☒	messages2	fa5610e05e3f	task4-messages2	8883:8883	1.46%	43 minutes ago	☒	:	☒
☐	☒	config-server	c1823d26ae83	task4-config-server	8881:8881	0.01%	43 minutes ago	☒	:	☒
☐	☒	facade	14c70c50083f	task4-facade	8880:8880	0.01%	43 minutes ago	☒	:	☒

Showing 15 items

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
14c70c50083f	task4-facade	"python facade-serv..."	44 minutes ago	Up 43 minutes	0.0.0.0:8880->8880/tcp	facade
4aab87cb2b42	task4-logging1	"python logging-serv..."	44 minutes ago	Up 44 minutes	0.0.0.0:50051->50051/tcp	logging1
c1823d26ae83	task4-config-server	"python config-serv..."	44 minutes ago	Up 44 minutes (healthy)	0.0.0.0:8881->8881/tcp	config-server
fa5610e05e3f	task4-messages2	"python messages-ser..."	44 minutes ago	Up 43 minutes	0.0.0.0:8883->8883/tcp	messages2
02c5f2fd9b59	task4-messages1	"python messages-ser..."	44 minutes ago	Up 43 minutes	0.0.0.0:8882->8882/tcp	messages1
947ad28fccb4	task4-logging3	"python logging-serv..."	44 minutes ago	Up 44 minutes	0.0.0.0:50053->50053/tcp	logging3
a84c0191cd9	task4-logging2	"python logging-serv..."	44 minutes ago	Up 44 minutes	0.0.0.0:50052->50052/tcp	logging2
986de52cb97	confluentinc/cp-kafka:7.3.0	"/etc/confluent/dock..."	47 minutes ago	Up 44 minutes (healthy)	9092/tcp, 0.0.0.0:9094->9094/tcp	kafka3
0aac83842a80	confluentinc/cp-kafka:7.3.0	"/etc/confluent/dock..."	47 minutes ago	Up 47 minutes (healthy)	0.0.0.0:9092->9092/tcp	kafka1
b690f4026ac	confluentinc/cp-kafka:7.3.0	"/etc/confluent/dock..."	47 minutes ago	Up 47 minutes (healthy)	9092/tcp, 0.0.0.0:9093->9093/tcp	kafka2
15d768b4da05	confluentinc/cp-zookeeper:7.3.0	"/etc/confluent/dock..."	47 minutes ago	Up 47 minutes (healthy)	2888/tcp, 0.0.0.0:2181->2181/tcp, 3888/tcp	zookeeper
af80e9803d0d	hazelcast/hazelcast:5.3	"hz start"	47 minutes ago	Up 47 minutes (healthy)	0.0.0.0:5702->5701/tcp	hazelcast2
0f95856a1c80	hazelcast/hazelcast:5.3	"hz start"	47 minutes ago	Up 47 minutes (healthy)	0.0.0.0:5701->5701/tcp	hazelcast1
04f578964217	hazelcast/hazelcast:5.3	"hz start"	47 minutes ago	Up 47 minutes (healthy)	0.0.0.0:5703->5701/tcp	hazelcast3

- Через HTTP POST записати 10 повідомлень *msg1-msg10* через **facade-service**

```
Windows PowerShell

PS D:\dist-sys\Task4> curl.exe -X POST http://localhost:8880/msg -d "content=msg1" {"content":"msg1","id":"a9360766-803c-4a8a-9b49-763a2f1899bd"}
PS D:\dist-sys\Task4> curl.exe -X POST http://localhost:8880/msg -d "content=msg2" {"content":"msg2","id":"49703404-e863-4d28-9af1-c64af0875f4f"}
PS D:\dist-sys\Task4> curl.exe -X POST http://localhost:8880/msg -d "content=msg3" {"content":"msg3","id":"a6708bec-97a3-4c48-846f-69e944e1bde4"}
PS D:\dist-sys\Task4> curl.exe -X POST http://localhost:8880/msg -d "content=msg4" {"content":"msg4","id":"3a2daf87-9e75-4de3-bd84-5cb155655e3c"}
PS D:\dist-sys\Task4> curl.exe -X POST http://localhost:8880/msg -d "content=msg5" {"content":"msg5","id":"87ef8e61-a40b-49ca-ac6c-6b3bde2f60d4"}
PS D:\dist-sys\Task4> curl.exe -X POST http://localhost:8880/msg -d "content=msg6" {"content":"msg6","id":"f8d6e821-4b94-4015-a3f0-e5dc488a3a71"}
PS D:\dist-sys\Task4> curl.exe -X POST http://localhost:8880/msg -d "content=msg7" {"content":"msg7","id":"7d22adc7-985a-4848-99cd-7d87d4f9b1c0"}
PS D:\dist-sys\Task4> curl.exe -X POST http://localhost:8880/msg -d "content=msg8" {"content":"msg8","id":"4994e447-5041-44e6-afda-eld9b37267b5"}
PS D:\dist-sys\Task4> curl.exe -X POST http://localhost:8880/msg -d "content=msg9" {"content":"msg9","id":"4c2f834d-aaa8-4e93-b28a-a1970c4abed3"}
PS D:\dist-sys\Task4> curl.exe -X POST http://localhost:8880/msg -d "content=msg10" {"content":"msg10","id":"092f5dcb-7cb7-4eaa-a271-7fc5088ad1c6"}
```

- Показати які повідомлення отримав кожен з екземплярів **logging-service** (це має бути видно у логах сервісу)

Logging1:

```
PS D:\dist-sys\Task4> docker logs logging1
INFO:hazelcast.lifecycle:HazelcastClient 5.5.0 is STARTING
INFO:hazelcast.lifecycle:HazelcastClient 5.5.0 is STARTED
INFO:hazelcast.connection:Trying to connect to Address(host=hazelcast2, port=5701)
INFO:hazelcast.lifecycle:HazelcastClient 5.5.0 is CONNECTED
INFO:hazelcast.connection:Authenticated with server Address(host=172.19.0.3, port=5701):2ca08421-6a6c-46b0-9124-94a0dalb62d5, server version: 5.3.8, local address: Address(host=172.19.0.8, port=41312)
INFO:hazelcast.cluster:
Members [3] {
  Member [172.19.0.2]:5701 - 9ff11f47-b5a7-4d20-886c-5ba29f4b103c
  Member [172.19.0.3]:5701 - 2ca08421-6a6c-46b0-9124-94a0dalb62d5
  Member [172.19.0.6]:5701 - 6429f436-c749-44fb-b84c-e06aa7af23d4
}
INFO:hazelcast.connection:Authenticated with server Address(host=172.19.0.2, port=5701):9ff11f47-b5a7-4d20-886c-5ba29f4b103c, server version: 5.3.8, local address: Address(host=172.19.0.8, port=44608)
INFO:hazelcast.connection:Authenticated with server Address(host=172.19.0.6, port=5701):6429f436-c749-44fb-b84c-e06aa7af23d4, server version: 5.3.8, local address: Address(host=172.19.0.8, port=47362)
INFO:hazelcast.client:Client started
INFO:_.main...[0948f83e55d3] Starting logging service on port 50051
* Serving Flask app 'logging-service'
* Debug mode: off
INFO:werkzeug:WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:50051
* Running on http://172.19.0.8:50051
INFO:werkzeug:Press CTRL-C to quit
INFO:_.main...[0948f83e55d3] Stored message ID 49703404-e863-4d28-9af1-c64af0875f4f
INFO:werkzeug:172.19.0.15 -- [16/Jun/2025 23:52:34] "POST /log HTTP/1.1" 201 -
INFO:_.main...[0948f83e55d3] Stored message ID a6708bec-97a3-4c48-846f-69e944e1bde4
INFO:werkzeug:172.19.0.15 -- [16/Jun/2025 23:52:36] "POST /log HTTP/1.1" 201 -
INFO:_.main...[0948f83e55d3] Stored message ID 092f5dcb-7cb7-4eaa-a271-7fc5088ad1c6
INFO:werkzeug:172.19.0.15 -- [16/Jun/2025 23:52:47] "POST /log HTTP/1.1" 201 -
PS D:\dist-sys\Task4> docker logs logging2
```

Logging2:

```

PS D:\dist-sys\Task4> docker logs logging2
INFO:hazelcast.lifecycle:HazelcastClient 5.5.0 is STARTING
INFO:hazelcast.lifecycle:HazelcastClient 5.5.0 is STARTED
INFO:hazelcast.connection:Trying to connect to Address(host=hazelcast2, port=5701)
INFO:hazelcast.lifecycle:HazelcastClient 5.5.0 is CONNECTED
INFO:hazelcast.connection:Authenticated with server Address(host=172.19.0.3, port=5701):2ca08421-6a6c-46b0-9124-94a0dalb62d5, server version: 5.3.8, local address: Address(host=172.19.0.7, port=47538)
INFO:hazelcast.cluster:
Members [3] {
  Member [172.19.0.2]:5701 - 9ff1f47-b5a7-4d20-886c-5ba29f4b103c
  Member [172.19.0.3]:5701 - 2ca08421-6a6c-46b0-9124-94a0dalb62d5
  Member [172.19.0.6]:5701 - 6429f436-c749-44fb-b84c-e86aa7af23d4
}
INFO:hazelcast.connection:Authenticated with server Address(host=172.19.0.2, port=5701):9ff1f47-b5a7-4d20-886c-5ba29f4b103c, server version: 5.3.8, local address: Address(host=172.19.0.7, port=48320)
INFO:hazelcast.connection:Authenticated with server Address(host=172.19.0.6, port=5701):6429f436-c749-44fb-b84c-e86aa7af23d4, server version: 5.3.8, local address: Address(host=172.19.0.7, port=59372)
INFO:hazelcast.client:Client started
INFO: _main...[4b5c0d070dc9] Starting logging service on port 50052
  * Serving Flask app 'logging-service'
  * Debug mode: off
INFO:werkzeug:WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
  * Running on all addresses (0.0.0.0)
  * Running on http://127.0.0.1:50052
  * Running on http://172.19.0.7:50052
INFO:werkzeug:Press CTRL+C to quit
INFO: _main...[4b5c0d070dc9] Stored message ID a9360766-803c-4a8a-9b49-763a2f1899bd
INFO:werkzeug:172.19.0.15 - - [16/Jun/2025 23:52:39] "POST /log HTTP/1.1" 201 -
INFO: _main...[4b5c0d070dc9] Stored message ID 3a2daf87-9e75-4de3-bd94-5cb155655e3c
INFO:werkzeug:172.19.0.15 - - [16/Jun/2025 23:52:37] "POST /log HTTP/1.1" 201 -
INFO: _main...[4b5c0d070dc9] Stored message ID 87ef8e61-a40b-49ca-ac6c-6b3bde2f66d4
INFO:werkzeug:172.19.0.15 - - [16/Jun/2025 23:52:39] "POST /log HTTP/1.1" 201 -
INFO: _main...[4b5c0d070dc9] Stored message ID f8d6e821-4b94-4015-a3f0-e5dc488a3a71
INFO:werkzeug:172.19.0.15 - - [16/Jun/2025 23:52:40] "POST /log HTTP/1.1" 201 -
INFO: _main...[4b5c0d070dc9] Stored message ID 7d22adc7-985a-4848-99cd-7d87d4f9b1c0
INFO:werkzeug:172.19.0.15 - - [16/Jun/2025 23:52:42] "POST /log HTTP/1.1" 201 -
INFO: _main...[4b5c0d070dc9] Stored message ID 4994e447-5041-44e6-afda-e1d9b37267b5
INFO:werkzeug:172.19.0.15 - - [16/Jun/2025 23:52:43] "POST /log HTTP/1.1" 201 -

```

Logging3:

```

PS D:\dist-sys\Task4> docker logs logging3
INFO:hazelcast.lifecycle:HazelcastClient 5.5.0 is STARTING
INFO:hazelcast.lifecycle:HazelcastClient 5.5.0 is STARTED
INFO:hazelcast.connection:Trying to connect to Address(host=hazelcast3, port=5701)
INFO:hazelcast.lifecycle:HazelcastClient 5.5.0 is CONNECTED
INFO:hazelcast.connection:Authenticated with server Address(host=172.19.0.2, port=5701):9ff1f47-b5a7-4d20-886c-5ba29f4b103c, server version: 5.3.8, local address: Address(host=172.19.0.9, port=40920)
INFO:hazelcast.cluster:
Members [3] {
  Member [172.19.0.2]:5701 - 9ff1f47-b5a7-4d20-886c-5ba29f4b103c
  Member [172.19.0.3]:5701 - 2ca08421-6a6c-46b0-9124-94a0dalb62d5
  Member [172.19.0.6]:5701 - 6429f436-c749-44fb-b84c-e86aa7af23d4
}
INFO:hazelcast.connection:Authenticated with server Address(host=172.19.0.3, port=5701):2ca08421-6a6c-46b0-9124-94a0dalb62d5, server version: 5.3.8, local address: Address(host=172.19.0.9, port=48448)
INFO:hazelcast.connection:Authenticated with server Address(host=172.19.0.6, port=5701):6429f436-c749-44fb-b84c-e86aa7af23d4, server version: 5.3.8, local address: Address(host=172.19.0.9, port=58290)
INFO:hazelcast.client:Client started
INFO: _main...[6c6ddef03c8] Starting logging service on port 50053
  * Serving Flask app 'logging-service'
  * Debug mode: off
INFO:werkzeug:WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
  * Running on all addresses (0.0.0.0)
  * Running on http://127.0.0.1:50053
  * Running on http://172.19.0.9:50053
INFO:werkzeug:Press CTRL+C to quit
INFO: _main...[6c6ddef03c8] Stored message ID 4c2f834f-aaa8-4e93-b28a-a1970c4bed3
INFO:werkzeug:172.19.0.15 - - [16/Jun/2025 23:52:45] "POST /log HTTP/1.1" 201 -
PS D:\dist-sys\Task4> |

```

- Показати які повідомлення отримав кожен з екземплярів **messages-service** (це має бути видно у логах сервісу)

Messages-service1:

```

PS D:\dist-sys\Task4> docker logs messages1
INFO: _main...[3219d11de5c0] Initializing Kafka consumer
INFO:kafka.conn:<BrokerConnection client_id=kafka-python-2.2.11, node_id=bootstrap-0 host=kafka3:9094 <connecting> [IPv4 ('172.19.0.10', 9094)]>: connecting to kafka3:9094 [['172.19.0.10', 9094] IPv4]
INFO:kafka.conn:<BrokerConnection client_id=kafka-python-2.2.11, node_id=bootstrap-0 host=kafka3:9094 <checking_api_versions_recv> [IPv4 ('172.19.0.10', 9094)]>: Broker version id notified as 2.6
INFO:kafka.conn:<BrokerConnection client_id=kafka-python-2.2.11, node_id=bootstrap-0 host=kafka3:9094 <connected> [IPv4 ('172.19.0.10', 9094)]>: Connection complete.
INFO:kafka.consumer.subscription.state:Updating subscribed topics to: ('msg-queue',)
INFO: _main...[3219d11de5c0] Kafka consumer ready
INFO: _main...[3219d11de5c0] Starting message consumption
INFO: _main...[3219d11de5c0] Starting message service on port 8882
  * Serving Flask app 'messages-service'
  * Debug mode: off
INFO:werkzeug:WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
  * Running on all addresses (0.0.0.0)
  * Running on http://127.0.0.1:8882
  * Running on http://172.19.0.14:8882
INFO:werkzeug:Press CTRL+C to quit
INFO:kafka.conn:<BrokerConnection client_id=kafka-python-2.2.11, node_id=1 host=kafka1:9092 <connecting> [IPv4 ('172.19.0.11', 9092)]>: connecting to kafka1:9092 [['172.19.0.11', 9092] IPv4]
INFO:kafka.conn:<BrokerConnection client_id=kafka-python-2.2.11, node_id=1 host=kafka1:9092 <connected> [IPv4 ('172.19.0.11', 9092)]>: Connection complete.
INFO:kafka.conn:<BrokerConnection client_id=kafka-python-2.2.11, node_id=bootstrap-0 host=kafka3:9094 <connected> [IPv4 ('172.19.0.10', 9094)]>: Closing connection.
INFO:kafka.cluster.coordinator:Coordinator for group/msg-group is BrokerMetadata(nodeId='coordinator-1', host='kafka1', port=9092, rack=None)
INFO:kafka.coordinator:Discovered coordinator coordinator-1 for group msg-group
INFO:kafka.coordinator.heartbeat:Starting new heartbeat thread
INFO:kafka.coordinator.consumer:Revoking previously assigned partitions set() for group msg-group
INFO:kafka.coordinator:Failed to join group msg-group: NodeNotReadyError: coordinator-1
INFO:kafka.conn:<BrokerConnection client_id=kafka-python-2.2.11, node_id=coordinator-1 host=kafka1:9092 <connecting> [IPv4 ('172.19.0.11', 9092)]>: connecting to kafka1:9092 [['172.19.0.11', 9092] IPv4]
INFO:kafka.conn:<BrokerConnection client_id=kafka-python-2.2.11, node_id=coordinator-1 host=kafka1:9092 <connected> [IPv4 ('172.19.0.11', 9092)]>: Connection complete.
INFO:kafka.coordinator:Received member id kafka-python-2.2.11-3c2788ba-b6db-4407-9d2a-01b734592874 for group msg-group; will retry join-group
INFO:kafka.coordinator:Failed to join group msg-group: [Error 79] MemberIdRequiredError
INFO:kafka.coordinator:(Re-)joining group msg-group
INFO:kafka.consumer.subscription.state:Updated partition assignment: {TopicPartition(topic='msg-queue', partition=2)}
INFO:kafka.coordinator.consumer:Setting newly assigned partitions {TopicPartition(topic='msg-queue', partition=2)} for group msg-group
INFO: _main...[3219d11de5c0] Consumed message ID a9360766-803c-4a8a-9b49-763a2f1899bd: msg1
INFO: _main...[3219d11de5c0] Consumed message ID 4994e447-5041-44e6-afda-e1d9b37267b5: msg0

```

Messages-service2:

```
PS D:\dist-sys\Task4> docker logs messages2
INFO:werkzeug:WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
INFO:werkzeug:Press CTRL+C to quit
INFO:kafka.com: <BrokerConnection client_id=kafka-python-2.2.11, node_id=bootstrap-1 host=kafka1:9092 <connecting> [IPv4 ('172.19.0.11', 9092)]>: connecting to kafka1:9092 [IPv4 ('172.19.0.11', 9092) IPv4]
INFO:kafka.com: <BrokerConnection client_id=kafka-python-2.2.11, node_id=bootstrap-1 host=kafka1:9092 <checking_api_versions_recv> [IPv4 ('172.19.0.11', 9092)]>: Broker version identified as 2.6
INFO:kafka.com: <BrokerConnection client_id=kafka-python-2.2.11, node_id=bootstrap-1 host=kafka1:9092 <connected> [IPv4 ('172.19.0.11', 9092)]>: Connection complete.
INFO:kafka.consumer.subscription_state: Updating subscribed topics to: ('msg-queue',)
INFO:werkzeug: [fc527436b361] Kafka consumer ready
INFO:werkzeug: [fc527436b361] Starting message consumption
INFO:werkzeug: [fc527436b361] Starting message service on port 8883
  * Serving Flask app 'messages-service'
  * Debug mode: off
INFO:werkzeug:WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
  * Running on all addresses (0.0.0.0)
  * Running on http://127.0.0.1:8883
  * Running on http://172.19.0.13:8883
INFO:kafka.com: <BrokerConnection client_id=kafka-python-2.2.11, node_id=2 host=kafka2:9093 <connecting> [IPv4 ('172.19.0.12', 9093)]>: connecting to kafka2:9093 [IPv4 ('172.19.0.12', 9093) IPv4]
INFO:kafka.com: <BrokerConnection client_id=kafka-python-2.2.11, node_id=2 host=kafka2:9093 <connected> [IPv4 ('172.19.0.12', 9093)]>: Connection complete.
INFO:kafka.cluster.coordinator: For group/msg-group is BrokerMetadata(nodeId='coordinator-1', host='kafka1', port=9092, rack=None)
INFO:kafka.coordinator: Discovered coordinator coordinator-1 for group msg-group
INFO:kafka.coordinator.heartbeat: Starting new heartbeat thread
INFO:kafka.coordinator.consumer: Revoking previously assigned partitions set() for group msg-group
INFO:kafka.coordinator: Failed to join group msg-group: MemberIdRequiredError: coordinator-1
INFO:kafka.com: <BrokerConnection client_id=kafka-python-2.2.11, node_id=coordinator-1 host=kafka1:9092 <connecting> [IPv4 ('172.19.0.11', 9092)]>: connecting to kafka1:9092 [IPv4 ('172.19.0.11', 9092) IPv4]
INFO:kafka.com: <BrokerConnection client_id=kafka-python-2.2.11, node_id=coordinator-1 host=kafka1:9092 <connected> [IPv4 ('172.19.0.11', 9092)]>: Connection complete.
INFO:kafka.coordinator: (Re-)joining group msg-group
INFO:kafka.coordinator: Received member id kafka-python-2.2.11-84bc312b-886b-4db6-8d55-661b652e3297 for group msg-group; will retry join-group
INFO:kafka.coordinator: Failed to join group msg-group: [Error 79] MemberIdRequiredError
INFO:kafka.com: <BrokerConnection client_id=kafka-python-2.2.11, node_id=coordinator-1 host=kafka1:9092 <connecting> [IPv4 ('172.19.0.11', 9092)]>: connecting to kafka1:9092 [IPv4 ('172.19.0.11', 9092) IPv4]
INFO:kafka.coordinator: Elected group leader -- performing partition assignments using range
INFO:kafka.coordinator: Successfully joined group msg-group <Generation 3 (member id: kafka-python-2.2.11-84bc312b-886b-4db6-8d55-661b652e3297, protocol: range)>
INFO:kafka.coordinator: Elected group leader -- performing partition assignments using range
INFO:kafka.consumer.subscription_state: Updated partition assignment: [TopicPartition(topic='msg-queue', partition=0), TopicPartition(topic='msg-queue', partition=1)]
INFO:kafka.coordinator.consumer: Setting newly assigned partitions [TopicPartition(topic='msg-queue', partition=1), TopicPartition(topic='msg-queue', partition=0)] for group msg-group
INFO:kafka.com: <BrokerConnection client_id=kafka-python-2.2.11, node_id=1 host=kafka1:9092 <connecting> [IPv4 ('172.19.0.11', 9092)]>: connecting to kafka1:9092 [IPv4 ('172.19.0.11', 9092) IPv4]
INFO:kafka.com: <BrokerConnection client_id=kafka-python-2.2.11, node_id=1 host=kafka1:9092 <connected> [IPv4 ('172.19.0.11', 9092)]>: Connection complete.
INFO:werkzeug: [fc527436b361] Consumed message ID 49783484-e863-4d28-9af1-c64af0875f4f: msg2
INFO:werkzeug: [fc527436b361] Consumed message ID a6788bec-97a3-4c48-846f-69e944e1bde4: msg3
INFO:werkzeug: [fc527436b361] Consumed message ID 3a2da8f7-9e75-4de3-bd84-5cb15565e3c: msg4
INFO:werkzeug: [fc527436b361] Consumed message ID 87e8fe61-24b0-49ca-acc6-6b3bde2f60d4: msg5
INFO:werkzeug: [fc527436b361] Consumed message ID f8d6e821-bb94-4015-a3f0-e5dc488a3a71: msg6
INFO:werkzeug: [fc527436b361] Consumed message ID 7d22adc7-985a-4848-99cd-7db7d4f9b1c0: msg7
INFO:werkzeug: [fc527436b361] Consumed message ID 4c2f834d-aaa8-4e93-b2ba-a1970c4abed3: msg9
INFO:werkzeug: [fc527436b361] Consumed message ID 892f9dcb-7cb7-4eaa-a271-7fc588aad1c6: msg10
PS D:\dist-sys\Task4>
```

- Декілька разів викликати HTTP GET на **facade-service** та отримати об'єднані дві множини повідомлень - це мають бути повідомлення з **logging-service** та **messages-service**

```
Windows PowerShell
PS D:\dist-sys\Task4> curl http://localhost:8880/msg

StatusCode      : 200
StatusDescription : OK
Content         : [{"combined_messages":[{"content":"msg1","id":"a9369766-883c-4a8a-9b49-763a2f1899bd"}, {"content":"msg8","id":"a994e447-5041-44e6-afda-e1d9b37267b5"}, {"content":"msg2","id":"49783484-e863-4d28-9af1-c64a...
RawContent      : HTTP/1.1 200 OK
                  Connection: close
                  Content-Length: 1738
                  Content-Type: application/json
                  Date: Tue, 17 Jun 2025 00:07:53 GMT
                  Server: Werkzeug/3.1.3 Python/3.10.18

                  [{"combined_messages":[{"content"...
Forms           : {}
Headers         : [{"Connection", "close"}, [{"Content-Length", 1738}, [{"Content-Type", "application/json"}, [{"Date", "Tue, 17 Jun 2025 00:07:53 GMT"}]...
Images          : {}
InputFields     : {}
Links           : {}
ParsedHtml      : mshtml.HTMLDocumentClass
RawContentLength : 1738

PS D:\dist-sys\Task4> curl http://localhost:8880/msg

StatusCode      : 200
StatusDescription : OK
Content         : [{"combined_messages":[{"content":"msg1","id":"a9369766-883c-4a8a-9b49-763a2f1899bd"}, {"content":"msg8","id":"a994e447-5041-44e6-afda-e1d9b37267b5"}, {"content":"msg2","id":"49783484-e863-4d28-9af1-c64a...
RawContent      : HTTP/1.1 200 OK
                  Connection: close
                  Content-Length: 1738
                  Content-Type: application/json
                  Date: Tue, 17 Jun 2025 00:07:54 GMT
                  Server: Werkzeug/3.1.3 Python/3.10.18

                  [{"combined_messages":[{"content"...
Forms           : {}
Headers         : [{"Connection", "close"}, [{"Content-Length", 1738}, [{"Content-Type", "application/json"}, [{"Date", "Tue, 17 Jun 2025 00:07:54 GMT"}]...
Images          : {}
InputFields     : {}
Links           : {}
ParsedHtml      : mshtml.HTMLDocumentClass
RawContentLength : 1738

PS D:\dist-sys\Task4> curl http://localhost:8880/msg

StatusCode      : 200
StatusDescription : OK
Content         : [{"combined_messages":[{"content":"msg1","id":"a9369766-883c-4a8a-9b49-763a2f1899bd"}, {"content":"msg8","id":"a994e447-5041-44e6-afda-e1d9b37267b5"}, {"content":"msg2","id":"49783484-e863-4d28-9af1-c64a...
RawContent      : HTTP/1.1 200 OK
                  Connection: close
                  Content-Length: 1738
                  Content-Type: application/json
                  Date: Tue, 17 Jun 2025 00:07:56 GMT
                  Server: Werkzeug/3.1.3 Python/3.10.18

                  [{"combined_messages":[{"content"...
Forms           : {}
Headers         : [{"Connection", "close"}, [{"Content-Length", 1738}, [{"Content-Type", "application/json"}, [{"Date", "Tue, 17 Jun 2025 00:07:56 GMT"}]...
Images          : {}
InputFields     : {}
Links           : {}
ParsedHtml      : mshtml.HTMLDocumentClass
RawContentLength : 1738

PS D:\dist-sys\Task4>
```

щоразу facade-service обирає випадкову копію messages-service і повертає її повідомлення разом із логами, які facade зберігав сам або через logging-service.

Перевірка відмовостійкості черги повідомлень

Класична черга повідомлень зберігає додані в неї повідомлення на своїй стороні, тому якщо її сервер раптово “впаде”, то повідомлення які містяться у черзі стануть недоступними та можуть взагалі втратитись.

Для запобігання цього, рекомендується налаштувати кластер для message queue, тобто запускати декілька серверів та налаштувати між ними реплікацію (чи backup для Hazelcast). Тепер якщо один сервер з кластеру message queue впаде, то повідомлення будуть доступні з його репліки (копії)

У даній роботі вам необхідно налаштувати такий кластер для Hazelcast (чи Kafka) та перевірити його відмовостійкість.

Для перевірки відмовостійкості вам необхідно буде перевірити, що не буде втрат повідомлень у випадку зупинки одного з серверів message queue.

Для цього:

- не запускайте (вимкніть) два екземпляри **messages-service** для того щоб повідомлення тимчасово не вичитувались та зберігались у message queue

```
PS D:\dist-sys\Task4> docker exec -it kafka1 /usr/bin/kafka-topics --bootstrap-server kafka1:9092 --describe --topic msg-queue
Topic: msg-queue      TopicId: 8b1nUSq-QP-5Q676zRAQWQ PartitionCount: 3      ReplicationFactor: 3      Configs:
Topic: msg-queue      Partition: 0      Leader: 2      Replicas: 2,3,1 Isr: 1,3,2
Topic: msg-queue      Partition: 1      Leader: 3      Replicas: 3,1,2 Isr: 1,3,2
Topic: msg-queue      Partition: 2      Leader: 1      Replicas: 1,2,3 Isr: 1,3,2
PS D:\dist-sys\Task4> |
```

Вимкнули два екземпляри **messages-service**

```
PS D:\dist-sys\Task4> docker stop messages1 messages2
messages1
messages2
```

☐	☒	logging2	4b5c0d070dc9	task4-logging2	50052:50052	0.89%	47 minutes ago			
☐	☐	messages1	3219d11de5c0	task4-messages1	8882:8882	0%	8 minutes ago			
☐	☒	logging3	6c66dfef83c8	task4-logging3	50053:50053	0.85%	47 minutes ago			
☐	☐	messages2	fc527436b361	task4-messages2	8883:8883	0%	7 minutes ago			
☐	☒	logging1	0948f83e55d3	task4-logging1	50051:50051	0.83%	47 minutes ago			
☐	☒	config-server	4346f500c5da	task4-config-server	8881:8881	1.46%	47 minutes ago			
☐	☒	facade	93aed286d966	task4-facade	8880:8880	0.03%	47 minutes ago			

Showing 15 items

- відправте 10 (можна і 100) повідомлень через **facade-service**

```
PS D:\dist-sys\Task4> curl.exe -X POST http://localhost:8880/msg -d "content=msg1"
{"content":"msg1","id":"835755bf-1723-4fdf-a8fa-f3a490885718"}
PS D:\dist-sys\Task4> curl.exe -X POST http://localhost:8880/msg -d "content=msg2"
{"content":"msg2","id":"959ea75c-3617-49bd-9ade-419109db9b42"}
PS D:\dist-sys\Task4> curl.exe -X POST http://localhost:8880/msg -d "content=msg3"
{"content":"msg3","id":"cf2a92ff-8de4-43f5-b67a-e5a51a1e2457"}
PS D:\dist-sys\Task4> curl.exe -X POST http://localhost:8880/msg -d "content=msg4"
{"content":"msg4","id":"397ce534-7ded-4e64-9be6-c6a13075fcf9"}
PS D:\dist-sys\Task4> curl.exe -X POST http://localhost:8880/msg -d "content=msg5"
{"content":"msg5","id":"2825fe73-230f-40bd-bc03-6833c5a96b58"}
PS D:\dist-sys\Task4> curl.exe -X POST http://localhost:8880/msg -d "content=msg6"
{"content":"msg6","id":"5d164086-fbef-4f1a-80f5-0d82c367749b"}
PS D:\dist-sys\Task4> curl.exe -X POST http://localhost:8880/msg -d "content=msg7"
{"content":"msg7","id":"784a5a8d-00da-40e6-8603-5cc764a591be"}
PS D:\dist-sys\Task4> curl.exe -X POST http://localhost:8880/msg -d "content=msg8"
{"content":"msg8","id":"675d628c-ce27-472d-ab6e-c509c8aa2d77"}
PS D:\dist-sys\Task4> curl.exe -X POST http://localhost:8880/msg -d "content=msg9"
{"content":"msg9","id":"a70b6cad-6bc1-4373-9679-870159fbc20e"}
PS D:\dist-sys\Task4> curl.exe -X POST http://localhost:8880/msg -d "content=msg10"
{"content":"msg10","id":"f25e3e20-a24e-4cc4-8957-b910e1fd621b"}
```

Логу facade:

```
INFO: kafka.conn->BrokerConnection Client id=kafka-python-producer-1 node_id=2 host=kafka2:9093 <connected> [IPv4 ('172.18.0.12', 9093)]: Connection complete.
INFO: _main_..Sent message to Kafka: {'id': '8a9d215-21b9-4027-a6f2-3df8708ac51f', 'txt': 'msg1'}
INFO: _main_..Fetching addresses for logging from http://config-server:8081
INFO: _main_..Received addresses for logging: [http://logging1:50051/logging', 'http://logging2:50052/logging', 'http://logging3:50053/logging']
INFO: _main_..Trying to send to http://logging3:50053/logging
INFO: _main_..Successfully sent to http://logging3:50053/logging
INFO: _main_..Sent message to Kafka: {'id': '241c75e8-8fed-4a9b-ba23-d5b12935fdac', 'txt': 'msg2'}
INFO: _main_..Fetching addresses for logging from http://config-server:8081
INFO: _main_..Received addresses for logging: [http://logging1:50051/logging', 'http://logging2:50052/logging', 'http://logging3:50053/logging']
INFO: _main_..Trying to send to http://logging2:50052/logging
INFO: _main_..Successfully sent to http://logging2:50052/logging
INFO: _main_..Sent message to Kafka: {'id': '3c0abb1c-8962-4d08-a696-93b64f6d98ac', 'txt': 'msg3'}
INFO: _main_..Fetching addresses for logging from http://config-server:8081
INFO: _main_..Received addresses for logging: [http://logging1:50051/logging', 'http://logging2:50052/logging', 'http://logging3:50053/logging']
INFO: _main_..Trying to send to http://logging1:50051/logging
INFO: _main_..Successfully sent to http://logging1:50051/logging
INFO: _main_..Sent message to Kafka: {'id': 'c2b2f358-b212-4df3-a63f-e8397596752b', 'txt': 'msg5'}
INFO: _main_..Fetching addresses for logging from http://config-server:8081
INFO: _main_..Received addresses for logging: [http://logging1:50051/logging', 'http://logging2:50052/logging', 'http://logging3:50053/logging']
INFO: _main_..Trying to send to http://logging2:50052/logging
INFO: _main_..Successfully sent to http://logging2:50052/logging
INFO: _main_..Sent message to Kafka: {'id': 'c651cd4b-b8ae-488a-b3b7-328d24eb5bd', 'txt': 'msg6'}
INFO: _main_..Fetching addresses for logging from http://config-server:8081
INFO: _main_..Received addresses for logging: [http://logging1:50051/logging', 'http://logging2:50052/logging', 'http://logging3:50053/logging']
INFO: _main_..Trying to send to http://logging1:50051/logging
INFO: _main_..Successfully sent to http://logging1:50051/logging
INFO: kafka.conn->BrokerConnection Client id=kafka-python-producer-1 node_id=2 host=kafka2:9093 <connecting> [IPv4 ('172.18.0.12', 9093)]: connecting to kafka2:9093 [IPv4 ('172.18.0.12', 9093)] IPv4
INFO: kafka.conn->BrokerConnection Client id=kafka-python-producer-1 node_id=2 host=kafka2:9093 <connected> [IPv4 ('172.18.0.12', 9093)]: Connection complete.
INFO: _main_..Sent message to Kafka: {'id': '6f717274-ca8e-4dd8-a3c7-7d799a2786de', 'txt': 'msg7'}
INFO: _main_..Fetching addresses for logging from http://config-server:8081
INFO: _main_..Received addresses for logging: [http://logging1:50051/logging', 'http://logging2:50052/logging', 'http://logging3:50053/logging']
INFO: _main_..Trying to send to http://logging3:50053/logging
INFO: _main_..Successfully sent to http://logging3:50053/logging
INFO: _main_..Sent message to Kafka: {'id': '6d133bf6-df02-4a4e-b1ad-dfd4798facad', 'txt': 'msg9'}
INFO: _main_..Fetching addresses for logging from http://config-server:8081
INFO: _main_..Received addresses for logging: [http://logging1:50051/logging', 'http://logging2:50052/logging', 'http://logging3:50053/logging']
INFO: _main_..Trying to send to http://logging1:50051/logging
INFO: _main_..Successfully sent to http://logging1:50051/logging
INFO: _main_..Sent message to Kafka: {'id': '878c329f-cc54-4f76-b55a-18f9fdb342bb', 'txt': 'msg10'}
INFO: _main_..Fetching addresses for logging from http://config-server:8081
INFO: _main_..Received addresses for logging: [http://logging1:50051/logging', 'http://logging2:50052/logging', 'http://logging3:50053/logging']
INFO: _main_..Trying to send to http://logging2:50052/logging
INFO: _main_..Successfully sent to http://logging2:50052/logging
PS D:\shared\dist-lab>
```

- вимкните один з серверів (message queue)
 - для Hazelcast це має бути сервер з меншою IP-адресою
 - для Kafka це має бути т.з. Leader

```
PS D:\dist-sys\Task4> docker stop kafka1
kafka1
PS D:\dist-sys\Task4> |
```

```
PS D:\dist-sys\Task4> docker start messages1 messages2
messages1
messages2
PS D:\dist-sys\Task4> |
```

Лідером є брокер2

```
PS D:\dist-sys\Task4> docker stop kafka2
kafka2
PS D:\dist-sys\Task4> docker exec -it kafka1 /usr/bin/kafka-topics --bootstrap-server kafka1:9092 --describe --topic msg-queue
Topic: msg-queue      TopicId: 8b1nUSq-QP-5Q676zRAQWQ PartitionCount: 3      ReplicationFactor: 3      Configs:
  Topic: msg-queue      Partition: 0      Leader: 3      Replicas: 2,3,1 Isr: 3,1
  Topic: msg-queue      Partition: 1      Leader: 3      Replicas: 3,1,2 Isr: 3,1
  Topic: msg-queue      Partition: 2      Leader: 1      Replicas: 1,2,3 Isr: 3,1
PS D:\dist-sys\Task4> |
```

Лідер тепер став брокер3

```
PS D:\dist-sys\Task4> docker exec -it kafka3 /usr/bin/kafka-topics --bootstrap-server kafka1:9092 --describe --topic msg-queue
Topic: msg-queue      TopicId: 8b1nUSq-QP-5Q676zRAQWQ PartitionCount: 3      ReplicationFactor: 3      Configs:
  Topic: msg-queue      Partition: 0      Leader: 3      Replicas: 2,3,1 Isr: 3,1
  Topic: msg-queue      Partition: 1      Leader: 3      Replicas: 3,1,2 Isr: 3,1
  Topic: msg-queue      Partition: 2      Leader: 1      Replicas: 1,2,3 Isr: 3,1
PS D:\dist-sys\Task4> |
```

- запустить два екземпляри **messages-service** і перевірте що вони отримали всі повідомлення з черги

Запускаємо заново:

```
PS D:\dist-sys\Task4> docker start messages1 messages2
messages1
messages2
```

Messages1:

```
INFO:kafka.conn:<BrokerConnection client_id=kafka-python-2.2.10, node_id=2 host=kafka2:9093 <connected> [IPv4 ('172.18.0.12', 9093)]: Connection complete.
INFO:kafka.consumer.fetcher:Resetting offset for partition TopicPartition(topic='messages-topic', partition=2) to offset 0.
INFO:___main___:[8678591d715a] Attempting to connect to Kafka
INFO:kafka.conn:<BrokerConnection client_id=kafka-python-2.2.10, node_id=bootstrap-2 host=kafka1:9092 <connecting> [IPv4 ('172.18.0.10', 9092)]: connecting to kafka1:9092 [[('172.18.0.10', 9092) IPv4]]
INFO:kafka.conn:<BrokerConnection client_id=kafka-python-2.2.10, node_id=bootstrap-2 host=kafka1:9092 <checking_api_versions_recv> [IPv4 ('172.18.0.10', 9092)]: Broker v
ersion identified as 2.6
INFO:kafka.conn:<BrokerConnection client_id=kafka-python-2.2.10, node_id=bootstrap-2 host=kafka1:9092 <connected> [IPv4 ('172.18.0.10', 9092)]: Connection complete.
INFO:kafka.consumer.subscription_state:Updating subscribed topics to: ('messages-topic',)
INFO:___main___:[8678591d715a] Connected to Kafka
INFO:___main___:[8678591d715a] Starting Kafka consumer
INFO:___main___:[8678591d715a] Starting on port 8882
* Serving Flask app 'messages-service'
* Debug mode: off
INFO:kafka.conn:<BrokerConnection client_id=kafka-python-2.2.10, node_id=2 host=kafka2:9093 <connecting> [IPv4 ('172.18.0.12', 9093)]: connecting to kafka2:9093 [[('172.18.0.12', 9093) IPv4]]
INFO:kafka.conn:<BrokerConnection client_id=kafka-python-2.2.10, node_id=2 host=kafka2:9093 <connected> [IPv4 ('172.18.0.12', 9093)]: Connection complete.
INFO:kafka.conn:<BrokerConnection client_id=kafka-python-2.2.10, node_id=bootstrap-2 host=kafka1:9092 <connected> [IPv4 ('172.18.0.10', 9092)]: Closing connection.
INFO:werkzeug:WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:8882
* Running on http://172.18.0.11:8882
INFO:werkzeug:Press CTRL+C to quit
INFO:kafka.cluster:Coordinator for group/messages-group is BrokerMetadata(nodeId='coordinator-2', host='kafka2', port=9093, rack=None)
INFO:kafka.coordinator.heartbeat:Starting new heartbeat thread
INFO:kafka.coordinator.consumer:Revoking previously assigned partitions set() for group messages-group
INFO:kafka.coordinator:Failed to join group messages-group: NodeNotReadyError: coordinator-2
INFO:kafka.conn:<BrokerConnection client_id=kafka-python-2.2.10, node_id=coordinator-2 host=kafka2:9093 <connecting> [IPv4 ('172.18.0.12', 9093)]: connecting to kafka2:9093 [[('172.18.0.12', 9093) IPv4]]
INFO:kafka.conn:<BrokerConnection client_id=kafka-python-2.2.10, node_id=coordinator-2 host=kafka2:9093 <connected> [IPv4 ('172.18.0.12', 9093)]: Connection complete.
INFO:kafka.coordinator:(Re-)joining group messages-group
INFO:kafka.coordinator:Failed to join group messages-group: [Error 79] MemberIdRequiredError
INFO:kafka.coordinator:(Re-)joining group messages-group
INFO:kafka.coordinator:Successfully joined group messages-group <Generation 3 (member_id: kafka-python-2.2.10-3e2228ba-4aca-41d6-99ce-ccf964c3ebfc, protocol: range)>
INFO:kafka.coordinator:Elected group leader -- performing partition assignments using range
INFO:kafka.consumer.subscription_state:Updated partition assignment: [TopicPartition(topic='messages-topic', partition=2)]
INFO:kafka.coordinator.consumer:Setting newly assigned partitions [TopicPartition(topic='messages-topic', partition=2)] for group messages-group
INFO:___main___:[8678591d715a] Received message: msg7 with ID 0f7122f4-ca48-4d48-a3c7-74799a270d6e
INFO:___main___:[8678591d715a] Received message: msg8 with ID fc392562-dc4f-44e4-adef-6bfef25f7932
INFO:___main___:[8678591d715a] Received message: msg9 with ID 0d12b20f-df92-4aa0-b14d-dfdf709faae6
```

Messages2:

```
version identified as 2.6
INFO:kafka.conn:<BrokerConnection client_id=kafka-python-2.2.10, node_id=bootstrap-2 host=kafka2:9093 <connected> [IPv4 ('172.18.0.12', 9093)]: Connection complete.
INFO:kafka.consumer.subscription_state:Updating subscribed topics to: ('messages-topic',)
INFO:___main___:[449e6fe707b6] Connected to Kafka
INFO:___main___:[449e6fe707b6] Starting Kafka consumer
INFO:___main___:[449e6fe707b6] Starting on port 8883
* Serving Flask app 'messages-service'
* Debug mode: off
INFO:kafka.conn:<BrokerConnection client_id=kafka-python-2.2.10, node_id=1 host=kafka1:9092 <connecting> [IPv4 ('172.18.0.10', 9092)]: connecting to kafka1:9092 [[('172.18.0.10', 9092) IPv4]]
INFO:kafka.conn:<BrokerConnection client_id=kafka-python-2.2.10, node_id=1 host=kafka1:9092 <connected> [IPv4 ('172.18.0.10', 9092)]: Connection complete.
INFO:kafka.conn:<BrokerConnection client_id=kafka-python-2.2.10, node_id=bootstrap-2 host=kafka2:9093 <connected> [IPv4 ('172.18.0.12', 9093)]: Closing connection.
INFO:werkzeug:WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:8883
* Running on http://172.18.0.13:8883
INFO:werkzeug:Press CTRL+C to quit
INFO:kafka.cluster:Coordinator for group/messages-group is BrokerMetadata(nodeId='coordinator-2', host='kafka2', port=9093, rack=None)
INFO:kafka.coordinator:Discovered coordinator coordinator-2 for group messages-group
INFO:kafka.coordinator.heartbeat:Starting new heartbeat thread
INFO:kafka.coordinator.consumer:Revoking previously assigned partitions set() for group messages-group
INFO:kafka.coordinator:Failed to join group messages-group: NodeNotReadyError: coordinator-2
INFO:kafka.conn:<BrokerConnection client_id=kafka-python-2.2.10, node_id=coordinator-2 host=kafka2:9093 <connecting> [IPv4 ('172.18.0.12', 9093)]: connecting to kafka2:9093 [[('172.18.0.12', 9093) IPv4]]
INFO:kafka.conn:<BrokerConnection client_id=kafka-python-2.2.10, node_id=coordinator-2 host=kafka2:9093 <connected> [IPv4 ('172.18.0.12', 9093)]: Connection complete.
INFO:kafka.coordinator:(Re-)joining group messages-group
INFO:kafka.coordinator:Received member id kafka-python-2.2.10-1807e6ae-a28f-4ac6-a3a7-17869bcbf4fa for group messages-group; will retry join-group
INFO:kafka.coordinator:Failed to join group messages-group: [Error 79] MemberIdRequiredError
INFO:kafka.coordinator:(Re-)joining group messages-group
INFO:kafka.coordinator:Successfully joined group messages-group <Generation 3 (member_id: kafka-python-2.2.10-1807e6ae-a28f-4ac6-a3a7-17869bcbf4fa, protocol: range)>
INFO:kafka.consumer.subscription_state:Updated partition assignment: [TopicPartition(topic='messages-topic', partition=0), TopicPartition(topic='messages-topic', partition=1)]
INFO:kafka.coordinator.consumer:Setting newly assigned partitions [TopicPartition(topic='messages-topic', partition=0), TopicPartition(topic='messages-topic', partition=1)] for group messages-group
INFO:kafka.conn:<BrokerConnection client_id=kafka-python-2.2.10, node_id=2 host=kafka2:9093 <connecting> [IPv4 ('172.18.0.12', 9093)]: connecting to kafka2:9093 [[('172.18.0.12', 9093) IPv4]]
INFO:kafka.conn:<BrokerConnection client_id=kafka-python-2.2.10, node_id=2 host=kafka2:9093 <connected> [IPv4 ('172.18.0.12', 9093)]: Connection complete.
INFO:___main___:[449e6fe707b6] Received message: msg3 with ID 35ceb81c-8962-43d0-a696-93b64f6d98ac
INFO:___main___:[449e6fe707b6] Received message: msg6 with ID d563c66b-b8a4-480a-83b7-328424eb6b6d
INFO:___main___:[449e6fe707b6] Received message: msg10 with ID 079c329f-cc5d-4f76-b58a-10f94db3a2bb
INFO:___main___:[449e6fe707b6] Received message: msg1 with ID 48a9d215-21b9-4027-a6f2-3df8705a5e1f
INFO:___main___:[449e6fe707b6] Received message: msg2 with ID 241c75e8-8fed-4a9b-ba23-d5b12935fdac
INFO:___main___:[449e6fe707b6] Received message: msg4 with ID 6a1886c1-8889-41b0-b198-75520f3462b9
```

GET-запити до facade:

```
PS D:\dist-sys\Task4> curl.exe http://localhost:8888/msg
{"combined_messages":[],"log_services":[{"instance":"6c66dfef83c8","messages":["msg6 (handled by 4b5c0d070dc9)","msg21 (handled by 0948f83e55d3)","msg1 (handled by 4b5c0d070dc9)","msg8 (handled by 6c66dfef83c8)","msg3 (handled by 0948f83e55d3)","msg20 (handled by 0948f83e55d3)","msg8 (handled by 4b5c0d070dc9)","msg23 (handled by 6c66dfef83c8)","msg3 (handled by 4b5c0d070dc9)","msg14 (handled by 0948f83e55d3)","msg11 (handled by 6c66dfef83c8)","msg9 (handled by 0948f83e55d3)","msg7 (handled by 4b5c0d070dc9)","msg17 (handled by 4b5c0d070dc9)","msg15 (handled by 0948f83e55d3)","msg10 (handled by 0948f83e55d3)","msg12 (handled by 4b5c0d070dc9)","msg2 (handled by 4b5c0d070dc9)","msg16 (handled by 4b5c0d070dc9)","msg5 (handled by 4b5c0d070dc9)","msg4 (handled by 4b5c0d070dc9)","msg24 (handled by 6c66dfef83c8)","msg25 (handled by 6c66dfef83c8)","msg4 (handled by 4b5c0d070dc9)","msg6 (handled by 6c66dfef83c8)","msg9 (handled by 6c66dfef83c8)","msg19 (handled by 4b5c0d070dc9)","msg5 (handled by 6c66dfef83c8)","msg2 (handled by 0948f83e55d3)","msg22 (handled by 4b5c0d070dc9)","msg13 (handled by 4b5c0d070dc9)","msg18 (handled by 6c66dfef83c8)","msg7 (handled by 0948f83e55d3)","msg1 (handled by 4b5c0d070dc9)","msg10 (handled by 6c66dfef83c8)"],"service":["http://logging3:50053/log"],"msg_services":[]}]}
PS D:\dist-sys\Task4> |
```